

Sourasish Ghosh

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SUMMARY

BS in Data Science and M.Sc. in Atmospheric Science with a strong foundation in Data Science and Physics. Skilled in **ML/DL, statistical modeling, and data engineering**, with hands-on experience building and deploying **end-to-end ML systems** (DNNs, Random Forests, Fuzzy models).

INTERNSHIP EXPERIENCE

Spaceborn <i>Machine Learning & Data Engineer</i>	Feb 2026 – April 2026
- Engineered a real-time computer vision and state-estimation pipeline for Drone OS, integrating YOLOv8n and MobileNetV2 ONNX with Unscented Kalman Filters (UKF) to fuse multi-modal sensor data (LIDAR, Ultrasonic, RPi Cam) for predictive tracking. - Designed and deployed an unsupervised machine learning engine using Isolation Forest to detect critical anomalies within high-frequency, multi-dimensional time-series data. - Architected a low-latency data streaming pipeline with Apache Kafka and Redis, optimizing high-throughput data ingestion via Avro serialization to power downstream ML models.	
Excelerate <i>Data Analyst Intern</i>	Dec 2025 – Jan 2026
- Conducted analysis on candidate hiring datasets as a Project Lead. - Used BI-tools, Python, and Excel for visualization and deep-dive analysis. - Identified key-findings in Yield management and Geographic dominance across 200 countries.	

PROJECTS

E-Commerce-Multimodal-RAG-System (Link)	2026
Multimodal RAG system with semantic search using Qdrant VectorDB , added text and image embeddings using OpenCLIP and Gemini 2.5-flash API. Backend with FastAPI , Pydantic with SlowAPI rate-limiter.	
Python-based Finance Backend System (Link)	2026
High-performance financial backend using FastAPI , SQLAlchemy , and PostgreSQL . Implemented RBAC for data security and Pydantic for type safety.	
Coughometer – DNN-based Cough Classifier (Link)	2025
End-to-end app built with DNN and FastAPI , containerized with Docker and deployed on Render . Trained on Coswara datasets.	
Thesis: Hybrid Fuzzy–Random Forest Model (Link)	2025
Achieved 88.6% accuracy on 24 years of Kolkata weather data using a hybrid Fuzzy Logic and Random Forest architecture.	

EDUCATION

Indian Institute of Technology, Madras <i>Bachelor of Science in Data Science</i>	2023 – 2027 <i>Chennai (Distance)</i>
CGPA: 7.73 (Concurrent with M.Sc.)	
University of Calcutta <i>Master of Science in Atmospheric Sciences</i>	2023 – 2025 <i>Kolkata</i>
CGPA: 8.155	
Chandernagore College (Burdwan University) <i>Bachelor of Science in Physics (Honours)</i>	2020 – 2023 <i>Chandernagore</i>
CGPA: 7.72	

TECHNICAL SKILLS

Languages	Python (OOP, typing, modules, async), SQL, Java, Bash, Fortran, L ^A T _E X
Machine Learning & AI	PyTorch, TensorFlow, Scikit-learn, OpenCLIP, LLM APIs (Claude,Gemini,OpenAI), AWS Bedrock, RAG, GenAI, LangChain, LangGraph, NLP, Spacy, MobileNet V2 ONNX, TensorRT, Agent Flows & Routing, Model Guardrails.
Data & Databases	Qdrant, ChromaDB, PostgreSQL, Redis, Apache Kafka, BigQuery, PySpark, Pandas, NumPy.
Cloud, DevOps & MLOps	Azure, GCP, AWS, Docker, Kubernetes (K8s), MLflow, Terraform, Git, GitHub Actions, CI/CD Pipelines, Pull Requests, Artifacts, Vercel, Render, VMs.
Frameworks & Tools	FastAPI, Flask, Uvicorn, pytest, unittest, Fixtures, Grafana, Prometheus, PowerBI, A/B Testing.
CERTIFICATIONS	

- Microsoft Azure Introduction to Cloud Infrastructure (AZ-900 series)